

Pre-Laboration Test 2

Name: _____ Process: _____

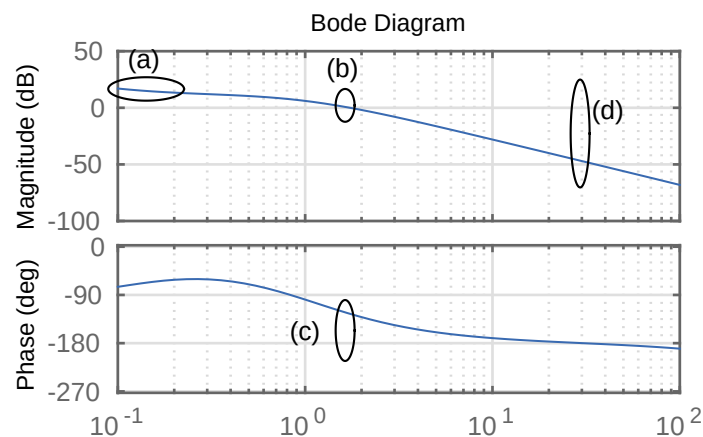
1. Which of the following contains complete information about how a system amplifies, attenuates and phase shifts signals?

- (a) The pole polynomial.
- (b) The frequency response function.
- (c) The stability criterion.
- (d) The controller structure.

2. Which one is the correct Laplace transform of $3\ddot{y} = 3\dot{y} - y + 4x$?

- (a) $Y(s) = \frac{4}{3s(s-1)+1}X(s)$
- (b) $Y(s) = \frac{s^2+4}{s(s-1)}X(s)$
- (c) $Y(s) = \frac{-3}{s(s-4)}X(s)$
- (d) $Y(s) = \frac{3s^2+1}{s^2}X(s)$

3. In which area is the phase margin to be read?



4. How many zeros does the following transfer function have?

$$G(s) = \frac{s-8}{s^2-4s-1}$$

- (a) 2
- (b) 3
- (c) 1
- (d) 5

5. What characterizes the resonance frequency ω_r ?

- (a) At this frequency the system becomes unstable.
- (b) It is the only frequency at which the amplification is positive.
- (c) The amplification at this frequency is above 3dB.
- (d) At this frequency there is a peak in amplification.

Answers:

1.	
2.	
3.	
4.	
5.	